

TABLE 1

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MT. GUNSON COPPER ORES, CHEMICAL ANALYSES (W/W%) BY CSR CENTRAL LABORATORY

Sample No.	Field No.	Sample	Total Cu	Total Fe	'Oxide' Cu	Chrysocolla Cu	Malachite Cu	Chalcocite Cu	Al ₂ O ₃	PO ₄ ³⁻	Insol. Cl
A 4000		Mt. Gunson Mill Feed 8.20 am, 15/7/1971	0.66	1.03	0.10	-	-	0.56	-	-	-
A 4001	2A	Lithic Sandstone West Lagoon	0.35	0.76	0.01	-	-	0.34	-	-	-
A 4005	4	Red Quartzite Ore West Lagoon	4.38	0.09	0.33	-	-	4.05	3.23	-	-
A 4006	5	Red Quartzite Ore West Lagoon	2.69	0.18	0.23	-	-	2.46	-	-	-
A 4007	5A	Lithic Sandstone immed., above Red Quartzite, West Lagoon	9.13	0.39	0.47	-	-	8.66	2.23	-	-
A 4008	6A	Lithic Sandstone West Lagoon	6.66	0.11	0.83	-	-	5.83	2.04	-	-
A 4009	6B	Red Quartzite West Lagoon	0.32	0.06	0.05	-	-	0.28	-	-	-
A 4010	6C	Red Quartzite West Lagoon	1.01	0.06	0.21	-	-	0.80	-	-	-
A 4012	9B	Red Quartzite Main Open Cut	8.34	-	-	6.41	0.87	N.D.	-	0.10	0.3
A 4013A	10	Metallurgical Test Area	10.31	-	-	8.51	0.77	N.D.	0.54	0.15	0.3
A 4013C	10	"	14.94	-	-	14.27	0.57	N.D.	-	0.05	0.4

N.D. : Not detected

'Oxide' Cu : Acid soluble copper (non-sulphide) and will include about half Cu O present.

Chrysocolla Cu : TBE float but included originally, malachite locked to crushed chrysocolla grains.

Malachite Cu : Determined from carbonate analysis as no alkaline earth carbonates were seen in thin-section.

Chalcocite Cu : Could include other sulphides present such as chalcopyrite, digenite, bornite and covellite and also half of any tenorite present.

TABLE 2

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MT. GUNSON COPPER ORES, CALCULATED MINERAL COMPOSITIONS (FROM CSR CENTRAL LABORATORY ANALYSES)

Sample No.	Field No.	Sample	Chalcocite Cu_2S	Malachite $\text{Cu}_2(\text{OH})_2\text{CO}_3$	Chrysocolla $\text{CuSiO}_3 \cdot 2\text{H}_2\text{O}$	Atacamite $\text{CuCl}_2 \cdot 3\text{Cu}(\text{OH})_2$	Kaolinite $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$	Calcium Phosphate $\text{Ca}_3(\text{PO}_4)_2$
A 4000	1	Mt. Gunson Mill Feed 8.20 am 15/7/1971	0.7	-	-	-	-	-
A 4001	2A	Lithic Sandstone West Lagoon	0.4	-	-	-	-	-
A 4005	4	Red Quartzite Ore West Lagoon	5.1	-	-	-	8.2	-
A 4006	5	Red Quartzite Ore West Lagoon	3.1	-	-	-	-	-
A 4007	5A	Lithic Sandstone above Red Quartzite, West Lagoon	10.8	-	-	-	5.6	-
A 4008	6A	Lithic Sandstone West Lagoon	7.3	-	-	-	5.2	-
A 4009	6B	Red Quartzite West Lagoon	0.4	-	-	-	-	-
A 4010	6C	Red Quartzite West Lagoon	1.0	-	-	-	-	-
A 4012	9B	Red Quartzite Main Open Cut	-	1.5	17.7	1.8	-	0.16
A 4013A	10A	Metallurgical Test Area	-	1.3	23.5	1.8	1.4	0.24
A 4013C	10C	Main Open Cut	-	1.0	39.4	2.4	-	0.08

000186

Mr. P.J. Curtis.

D.G. Tonkin.

23rd. October, 1973.

PANDURRA THIN SECTIONS.

A further 27 thin sections have been sent to you for study.
they are described briefly as follows:

- 7145 P.35 : P/Flaggy, micaceous siltstone; from "Pandurra".? Tregolana Shale.
- 7147 P.39 : Ironstone (conglomeratic). Anomalous copper.
- BPB? P.39A: P/ Whyalla Sandstone; ? basal conglomerate; silicified; from "Illeroo".
- 7157 P.40 : Sandstone. whyalla or Corraberra? from "Illeroo".
- 7159 P.41A: P/ Sandstone similar to P.40, from "Lincoln Gap".
- BPB? 7162 P.42 : Whyalla Sandstone, basal, silicified.
- P.43 : P/ Pandurra Formation quartzite, from regolith.
- 7163 P.44 : Silcrete, from Whyalla Sandstone, basal section.
- P.45 : P/ Gravel - believed to be basal portion of Whyalla Sandstone Unit.
- Sample finer grained than usual. Compare with P.39A.
- 7165 P.46 : P/ Whyalla Sandstone, overlying Gawler Volcanics.
- P.48 : P/ Pandurra Formation, from below the pre-Marinoan Zone. of silicification.
- BPB? P.49 : Silcreted gravel or conglomerate.
- 7166 P.49A: P/ Whyalla Sandstone, silicified and containing Pandurra pebbles.
- P.50B: P/ Looks like Pandurra Fm., outcrops like Whyalla Sandstone.
- P.51 : P/ Sandstone (Whyalla or Corraberra?) from Tent Hill crater.
- P.51B: P/ Pandurra Fm. Pink quartzite. Boulder in Whyalla Sandstone.
- 7173 P.52 : P/ ? Woocalla Dolomite. Manganiferous.
- 7175 P.53 : P/ Whyalla Sandstone, containing chips of Gawler Volcanics.
- 7181 P.56 : P/ Whyalla Sandstone, silicified.

P.57A: ✓ Silicified conglomerate; overlies Pandurra Formation.

Believed to have undergone pre-Marinoan silicification along
with upper section of Pandurra Formation.

P.58 : ✓ Pandurra Formation grit. Unsilicified.

7192 P.60 : P ✓ Whyalla Sandstone. Silicified lithic sandstone.

P.61 : ²
Silcret? Contains fibrous laths.

7199 P.65 : P ✓? Whyalla Sandstone. Clay-rich layers. Similar to (P.40, P.41A.

P.65A: ✓ ? Whyalla Sandstone. Similar to P.65.

P.66 : P ✓ Sandstone (Whyalla or Corraberra?).

P.67 : / Sandstone - please identify.

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PACMINEX PTY LIMITED

PETROGRAPHY OF PANDURRA

THIN SECTIONS. II.

SYDNEY

January, 1974

P.J. CURTIS

c.c. Library

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1. INTRODUCTION

A further 27 thin sections were sent to Head Office for stratigraphic identification and petrographic study. The field identification as is usual heads each description.

A series of specimens, P 2 to P 34 have already been examined and reported (Ref. 1).

2. SUMMARY

- P 35 A micaceous arenite with a similar mineral composition to some Tregolana Shale specimens seen by the writer.
- P 39 Probably a ferruginous variety of Whyalla Sandstone.
- P 39A Whyalla Sandstone as suggested by rounding of quartz grains and presence of lithic cherts.
- P 40 Corraberra Sandstone as indicated by abundance of 'flat' quartz grains, appreciable micaceous chert and heavy minerals present.
- P 41A Corraberra Sandstone as shown by appreciable 'flat' quartz grains and micaceous chert and abundant heavy minerals present.
- P 42 Whyalla Sandstone with sparse matrix of carbonate and chert.
- P 43 Pandurra Formation with a 'quartz-mud' clast.
- P 44 Derived Whyalla Sandstone mineral grains associated with chalcedonic fragments in a largely opaque cherty cement.

- P 45 Mainly cherty gravel with some 'pebbles' showing definite affinities with basal Whyalla Sandstone. Similar quartz grain shapes to P 39A but P 45 'pebble' is finer grained and part silicified.
- P 46 A peculiar sandstone containing volcanic feldspathic grains besides even-grained quartz and a tendency to flat quartz grains. These properties with the variety of heavy minerals present link the rock with Corraberra Sandstone.
- P 48 Pandurra Formation, largely silicified.
- P 49 A secondary silicified gravel which contains abundant coarse fragments of Pandurra Formation in a matrix of Whyalla Sandstone and rock material derived from other sources.
- P 49A Whyalla Sandstone - silicified, with inclusions of erratics containing fine recrystallised slightly ferruginous chert not obviously of Pandurra Formation origin, but this is the most likely source.
- P 50B Probably Whyalla Sandstone as there is a degree of roundness of much of the quartz present, a tendency to a two population distribution of quartz grains and appreciable ferruginous cherts present.
- P 51 Corraberra Sandstone. Strained quartz, many small flattened grains of quartz, the presence of appreciable micaceous chert and appreciable leucoxene are factors for affiliation with Corraberra Sandstone rather than with Whyalla Sandstone.
- P 51B Poor sorting of subangular quartz grains, lack of ferruginous chert and lack of heavy minerals favour classifying the rock as a member of the Pandurra Formation.

- P 52 Woocalla Dolomite at Mt. Gunson undoubtedly contains manganese as part of the dolomite molecule which oxidises during weathering, the manganese being released as dioxide and depositing in a diffuse fashion to give the rock a dark appearance. In this specimen, however, the manganese is deposited in and around fractures.
- P 53 A Whyalla Sandstone with abundant volcanic clasts. Silicification which resulted in quartz overgrowths has made round quartz grains take on a more angular aspect.
- P 56 Whyalla Sandstone containing quite spherical quartz, volcanic (basaltic ?) clasts, with clay and cherty rock cement in channelways and vugs.
- P 57A A 'pebbly-sand' type of conglomerate of largely Pandurra Formation origin which has been subsequently silicified.
- P 58 A specimen of silicified Pandurra Formation apparently transitional to Whyalla Sandstone.
- P 60 A silicified non-standard (ref. Mount Gunson) Whyalla Sandstone showing rough sorting of grains. The typical large round quartz grains and abundant volcanic cherts put the identity of the rock beyond doubt.
- P 61 A recrystallised quartzite containing ferruginous cherts including banded iron cherts as clasts; probably the latter are the fibrous laths which were noted in the field notes.
- P 65 Corraberra Sandstone with derived Whyalla Sandstone material as round lithic fragments and some spherical quartz rock fragments.

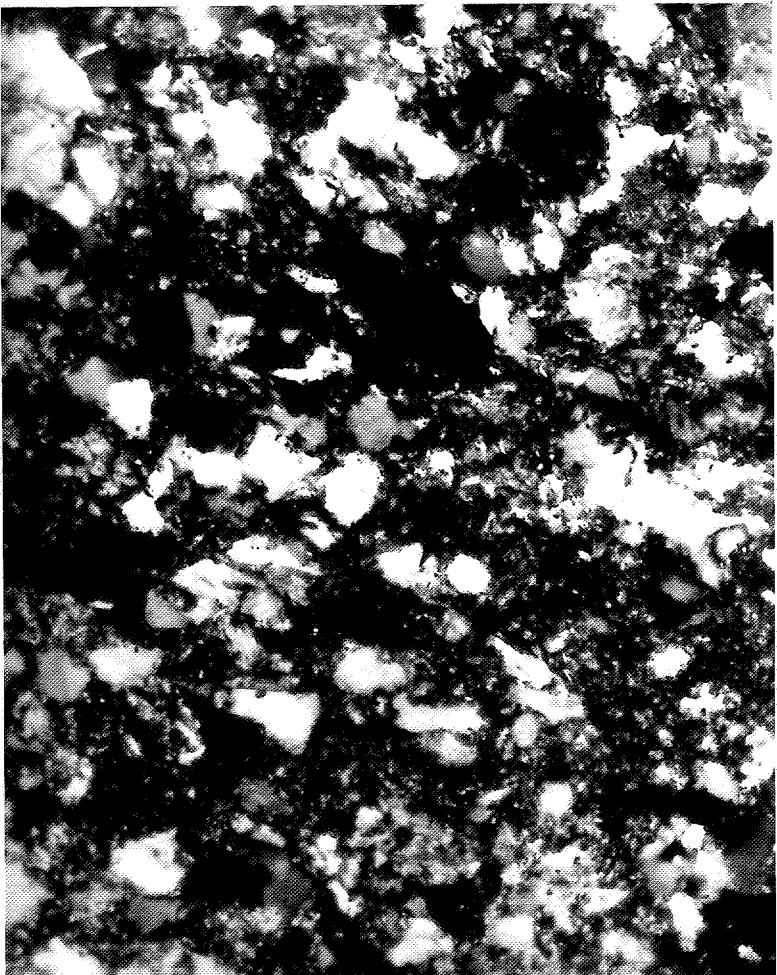


Figure 1

A micaceous arenite, possibly Tregolana Shale. P35
Quartz (flat grey or white colour, angular to
rounded grains).

Carbonate (Dark grey, lightly speckled grey and
white grains. Where limonite stained, the
grains are almost uniform black).
Limonite (black).

Sericite and chlorite (speckled grey and white
patches or books of ragged white grains).

Crossed nicols

Magnification X 105

- P 65A The specimen resembles both Whyalla and Corraberra Sandstone. With an abundance of cherty lithic grains, and few heavy mineral grains it is similar to Whyalla Sandstone; but is like Corraberra Sandstone with an appreciable number of quartz grains which are somewhat flattened. Generally there are some resemblance in quartz fabric to Corraberra Sandstone, specimen A 4223, Mt. Gunson.
- P 66 Looks to be transitional between Corraberra Sandstone and Whyalla Sandstone but with appreciable numbers of flat quartz grains and heavy minerals present, the rock is described as Corraberra Sandstone with derived Whyalla material (large spherical quartz grains and other large rounded mineral grains).
- P 67 Pandurra Formation resembling Pandurra Quartzite from some Mt. Gunson Main Open Cut specimens of angular to subangular quartz erratics in a fine quartz matrix.

3. MICROSCOPICAL INVESTIGATION

- P 35 : Flaggy, micaceous siltstone; from 'Pandurra' ?
Tregolana Shale (Figure 1).

A sericitic siltstone with banding emphasised by limonite staining. In thin-section the rock consists of long fibrous sericite laths as aggregates and short-staple laths in books which are unoriented parallel growths, alternating with anhedral quartz (approx. 0.08 mm) which is fairly even-grained, chlorite and patchy limonite as minute granules which are mainly intergrown with the micaceous minerals; and a little granular carbonate.

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Diagenetic recrystallisation of quartz has resulted in angular to subangular grains and mosaic patches. Similarly, sericite occurs in pockets and is interstitial to patchy quartz mosaics. Patchy anhedral albite-twinned albite plagioclase grains of similar grain size to quartz mosaics are seen.

Modal mineral percent volume composition :-

Quartz	79
Sericite	10
Chlorite	1
Carbonate	2
Albite	1
Limonite	7

There are similarities in composition with A4239 (DH 3-6-2D, 5.0 m), Mystery Area, Mt. Gunson, Tregolana, Shale; the rock (P 35) is however coarse grained with a median quartz of sandstone grain size (Pettijohn, Ref. 3). No resemblance to P 6B specimen from Pandurra, also thought to be Tregolana Shale, was noted.

The rock could be described as a micaceous arenite (Ref. 4) with a similar mineral composition to some Tregolana Shale specimens examined.

P 39 : Ironstone (conglomeratic). Anomalous copper.

In hand specimen the rock has the superficial appearance of a lateritic surface deposit.

In thin section, the rock consists of about 70 volume percent spherical quartz mineral grains and minor quartz rock fragments, varying between 0.05 and 6.0 mm, median 0.5 mm. Some round ferruginous cherts are also seen and the larger rock fragments may be of irregular Pandurra Quartzite with subangular constituent quartz

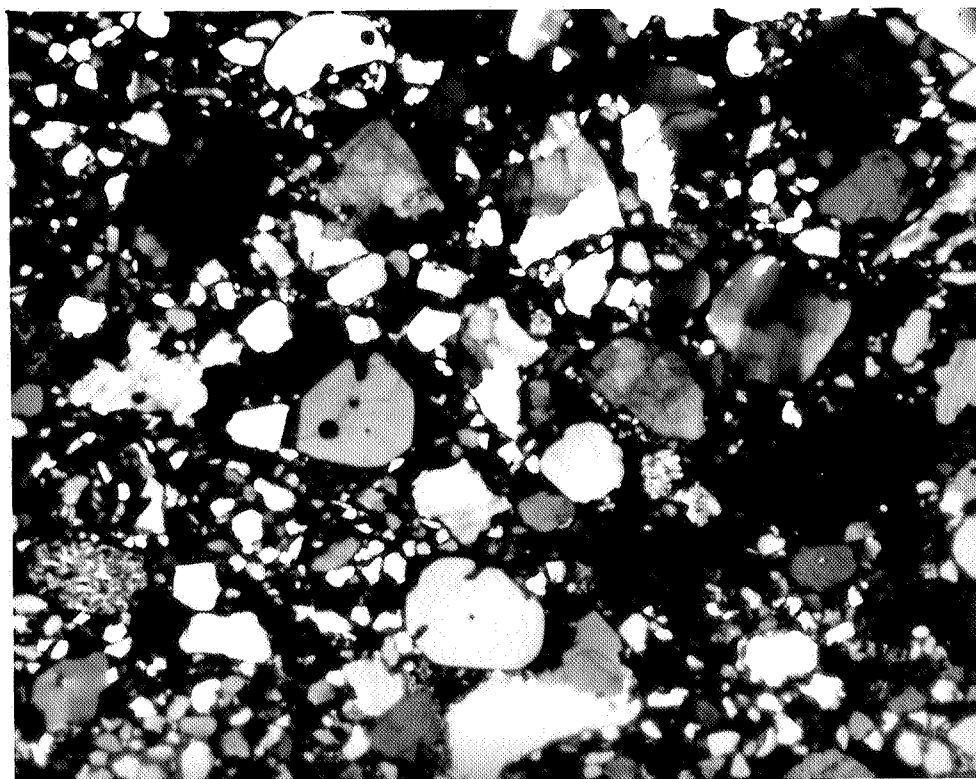


Figure 2

A probable ferruginous variety of Whyalla Sandstone. P39A
Quartz mineral grains (flat grey and white colour)
are rounded or subangular.
Quartz rock fragments (alternating grey and white
areas) vary from rounded to angular forms.
Chert (speckled grey and white). One grain occurs
near left bottom corner.
Clay mineral with associated limonite (dark back-
ground).

Crossed nicols

Magnification X 14

grains and a cherty cement. The rock matrix is entirely of limonite. The proportion of cherts (about half of these are ferruginous) is 2 to 3 volume percent. The rock is possibly a ferruginous variety of Whyalla Sandstone where the original clay matrix has been replaced by iron from percolating solutions.

P 39A : Whyalla Sandstone; ? basal conglomerate; silicified from 'Illeroo' (Figure 2).

The hand specimen shows mainly round quartz grains in a white clay matrix. Many specks of atacamite (?) occur at one end of the specimen.

The thin-section indicates rather more than half the larger grains (0.5 mm +) to be rounded; the remainder are subangular or a combination of subangular and rounded forms. More than half of the larger quartz grains are mineral grains. Angular quartz-gneiss grains occur (2 volume percent). Rounded cherts which are not ferruginous (vary between 0.5 to 1.0 mm), are present to 1 volume percent. Smaller quartz grains (<0.5 mm) are mainly mineral grains and are subangular. The matrix consists of fine chert and patchy limonite. The chert has made inroads on and has replaced some quartz grains along fractures or along margins of the grains. Submerged and almost obscured in the cherty matrix are seen some ferruginous rounded cherts and volcanic tachylyte with some fragments containing abundant limonite pseudomorphs of magnetite; the volcanic and ferruginous cherts total to possibly 3 volume percent.

Rounding of many quartz grains, presence of ferruginous cherts including volcanic fragments suggests this rock to be Whyalla Sandstone. Feldspar is apparently absent, and the rock appears to be completely silicified, although some 'cherts' contain sericitic material.

P 40 : Sandstone, Whyalla or Corraberra ? From Illeroo.

Bleached-looking sandstone with clay matrix (?) and traces of atacamite as greenish flakes which dissolve slowly in 1:1 hydrochloric acid.

There is a tendency for two generations of quartz grain-size to occur. The larger grains are mainly rounded and subangular with appreciable flat grains with rounded corners (overall variation 0.24 to 0.72 mm). These are nearly all quartz grains and there are few rock fragments. The smaller size mineral grain population is more subangular than round and varies mainly between 0.08 and 0.20 mm of quartz mineral grains, ferruginous cherts, volcanic 'cherts', and fresh feldspar (microcline > plagioclase). Illite occurs interstitially as a film between quartz grains and in some vugs.

The chert, (mainly micaceous/ferruginous) including silicified volcanic fragments, is about 7 volume percent of the rock. Albite and microcline occur together at about 1 volume percent. Bundles of greenish biotite and squat blue-green tourmaline (approx. 0.1 x 0.05 mm) are present as minor quantities. Possible monazite is scattered around the section as squat yellow sections; also scattered magnetite euhedra, blue-green tourmaline and zircon. The heavy minerals total just over 1 volume percent.

Appreciable flattened quartz grains, abundant micaceous chert and heavy mineral content are factors which classify this rock as Corraberra Sandstone.

P 41A : Sandstone similar to P 40, from Lincoln Gap.

Pale sandstone with a pebble band of spherical quartz grains.

The main sandstone consists of near spherical to flat and subangular quartz grains 0.2 to 1.0 mm, coarser chert grains 1.0 to 3.0 mm, mainly ferruginous and micaceous (with limonite as homogeneous specks in chert and fine mica matrix). The larger grains occur in a fine subangular quartz (approx. 0.1 mm) matrix. The abundance of ferruginous/micaceous chert (5 volume percent of section), two populations of clastics and roundness of some of the larger quartz grains, (Whyalla Sandstone; and flatness of many quartz grains and appreciable detrital rutile, zircon and tourmaline to about 1 percent, (Corraberra Sandstone); suggest this rock to be Corraberra Sandstone with derived Whyalla Sandstone material. Narrow channels between all quartz grains contain kaolinite flakes with rather less illite totalling to about 10 volume percent.

A fine grain sandstone with 10 volume percent quartz (largely mineral grains, some rock fragments) as coarse clastic material.

The pebble band is a sediment horizon in the rock consisting of large round quartz rock fragments and chert varying between 0.5 and 4.0 mm.

P 42 : Whyalla Sandstone, basal, silicified.

A pale sandstone of rounded quartz grains, moderate limonite-stained matrix and prominent iron-stained chert grains.

In thin section the quartz grains are seen to be somewhat rounded although secondary silicification has given some originally round grains a subangular shape, the quartz grains together forming quartz mosaics. Quartz grain size varies from 0.05 to 1.3 mm though most grains (including chert), approximate to median grain size of

0.6 mm. Carbonate seems to be replacing fine chert as the matrix (approx. 7 volume percent) between the quartz and primary chert grains. Ferruginous primary chert corresponds to 3 volume percent of rock. The rock is Whyalla Sandstone with sparse carbonate and chert matrix.

P 43 : Pandurra Formation Quartzite with regolith. ?
Boulder

Under low power microscopy the rock appears to be an irregular Pandurra Quartzite with pinkish quartz, clay matrix and 'regolith' consisting of rounded quartz grains with limonite cement.

In thin-section is seen no distinct boundary contact between the 'regolith' of coarser rounded quartz mineral grains (to 2.5 mm but variable and median 0.6 mm) and rock fragments and chert (to 3 mm); with a finer grained zone of mainly quartz mineral grains (approx. 0.15 mm but up to 2 mm) surrounding the 'regolith'. Below the regolith in the main host rock, banding of fine quartz grains (approx 0.2 mm) and unsorted (0.3 to 2 mm) bands are seen. The difference in cementing material is noted, that of the 'regolith' being mainly amorphous limonite and that of the host quartzite being largely silica but with occasional pockets of kaolinite platelets of similar size to the surrounding quartz grains. The quartz grains of the host quartzite are subangular and fit together as mosaics due to the normal formation of quartz overgrowths.

Within the 'regolith' the grains are also locally silicified and form mosaics, though blind channels and pockets of limonite occur and more than half of the grains show a degree of roundness. Where present the subangular grains were rounded before silicification as indicated by internal trains of fluid inclusions and limonite particles. Some small rounded cherts (< 1 volume percent of the 'regolith') - not ferruginous, are also seen.

No heavy minerals are seen.

Lack of distinct boundaries between the main rock and 'regolith' shows the latter to have been a possible unconsolidated aggregate of quartz held together by limonitic-mud, the resulting mud-ball being washed into the sedimentary environment by flood waters.

No regolith 'sensu stricto' is identifiable.

The host rock is identified as Pandurra Quartzite.

P 44 : Silcrete from Whyalla Sandstone, basal section

Some round quartz mineral grains are seen but the specimen consists mainly of subangular to angular chalcedonic fragments in a siliceous matrix as seen in hand specimen.

In thin-section each fragment of chalcedony (0.1 to 4.0 mm) consists mainly of aggregates of radial structures of silica; otherwise of cherty aggregates with quartz fragments. Rounded and subangular quartz mineral grains and rock fragments (median 1.8 mm) and fewer subangular grains, one containing a cherty volcanic fragment of tachylitic glass are seen. Ferruginous cherts as seen normally in Whyalla Sandstone are also present.

Ratio of rounded quartz to grains showing some angularity is about 1:3.

Chalcedony is more abundant than fine chert and the ratio of round grains to grains of angular aspects here is 1:20.



Figure 3

Mainly cherty gravel with some Whyalla Sandstone pebbles (one is shown at centre of photograph). P45

Quartz grains (grey, white and black).

Chert grains (speckled or mottled, grey, white or dark).

Left bottom of photograph is part of quartz rock fragment.

Left top of photograph and top centre are part of limonitic cherty grains.

Crossed nicols

Magnification X 15

Modal mineral composition :-

Quartz mineral grains	9
Quartz rock fragments	2
Chalcedony (much fragmented in places)	13
Chert (ferruginous and volcanic)	6
Siliceous matrix	70

The matrix is amorphous-cherty with scattered limonite particles.

The presence of some ferruginous chert, volcanic chert and round quartz show an affinity with normal Whyalla Sandstone. The presence of abundant silica-cement matrix and abundant fragmental chalcedony shows this rock to be a duricrust of mixed origin, with a proportion of Whyalla Sandstone grains.

P 45 : Gravel, believed to be basal Whyalla Sandstone Unit. (Figure 3). Sample finer grained than usual. Compare with P 39A. *(Loose gravel scooped off surface)*

The pebbles (approx. 6 mm diameter) are largely of chert or rounded quartz rock fragments with few quartz mineral grains. The cherts are round, subangular and flat with round edges. About half the chert grains are typical Whyalla ferruginous cherts and some contain devitrified volcanic glass and silicified feldspathic matter.

Composite percentage :-

	volume percent
Quartz mineral grains	2
Quartz rock fragments	25
Clear chert	21
Ferruginous cherts	33
'Volcanic' cherts	13
Shale (showing bedding and fine quartz particles)	6

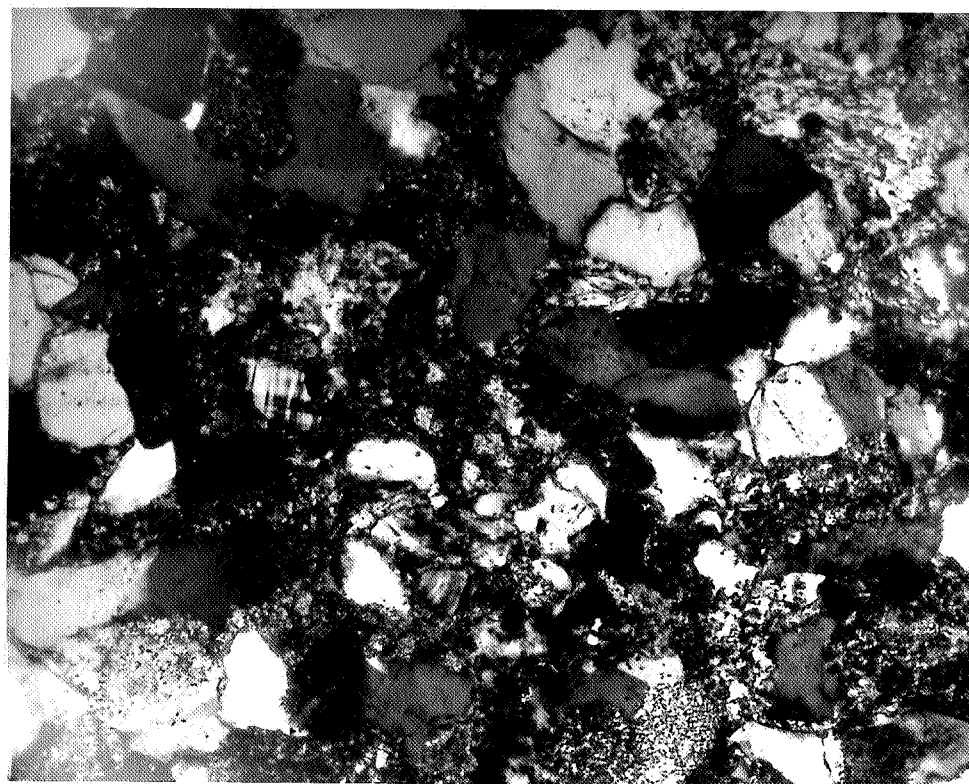


Figure 4

Corraberra Sandstone. P46

Quartz grains (flat grey and white colours) are subangular or flat shaped.

Micaceous and cherty matrix (speckled black and white areas between quartz grains).

Volcanic cherts (also speckled black and white but are darker grains).

Microcline grain (left of centre, twinned).

Crossed nicols

Magnification X 104

In specimen P 39A most clasts are quartz mineral grains with fewer rock fragments and more than half these grains are rounded. In P 45 about two-thirds of the 'clasts' are chert. The ferruginous cherts and volcanic cherts show some resemblance to those seen in P 39A.

If P 45 is derived from the Whyalla Sandstone then there has been some natural physical differentiation of liberated particles to give a chert-rich concentrate at point of sampling. However, there is no doubt that many of these pebbles come from a similar source to Whyalla Sandstone.

P 46 : Whyalla Sandstone, overlying Gawler Volcanics (Figure 4)

A very fine-grained sandstone with carbonate cement and containing patchy oxide-copper mineralisation.

In thin-section the quartz grains are more or less equigranular, subangular with an abundance of flat grains (0.05 to 0.24, median 0.16 mm) and many show replacement of grain margins by the carbonate matrix. In fact the matrix is largely brownish chloritic clay with patchy illite and carbonate. The chloritic clay platelets commonly occupy the centres of vugs; illite and carbonate, mixed or separately occupy narrow or broad zones flanking the quartz grains; however chloritic clay platelets also flank quartz grains locally. Chloritic clay is associated with leucoxene which occurs as granule aggregates. Occasional grains of fresh albite (one percent of rock) and a little limonitised magnetite are seen as scarce components; also large (0.3 x 0.8 mm) flakes of muscovite; less than one percent volume of rock of cherty tachylite; and large (0.10 mm) chunky tourmaline (olive green) grains. The abundance of chloritic clay, scattered leucoxene aggregates, and feldspar and tachylitic chert fragments may indicate derivation of matrix from an igneous rock source.

Modal composition of rock :-

	Volume percent
Quartz	66
Chert (including much micaceous chert)	2
Volcanic fragments	1
Carbonate	3
Chloritic clay	20
Illite/muscovite	5
Leucoxene	2
Plagioclase feldspar	1
Tourmaline, topaz and sphene	<1

The fine grain, predominance of equigranular quartz including many flattened grains scattered opaques and appreciable heavy minerals make this rock more akin to Corraberra Sandstone (comparable with specimen A4224, Mt. Gunson).

P 48 : Pandurra Formation, from below the pre-Marinoan zone of silicification.

In hand specimen, subangular grains of quartz are moderately sorted, with pebbles several millimetres diameter forming the 'bottom' end of the specimen. Malachite is seen as traces in the interstices of finer sizes of quartz grains.

Thin-section shows very little variation between mainly quartz mineral grains which are largely subangular to sub-rounded and form a mosaic. The variation in grain size is between 0.1 and 1.0 mm with a median grain size of 0.3 mm. At the coarser pebbly end of the section the grain size abruptly rises to 3.5 mm consisting of rounded and flattened grains. Those grains between 1.2 and 3.5 mm constitute about 30 volume percent of grains present at the pebbly end. The remainder of grains present are of similar size range to quartz grains at the finer end. Thus, the larger quartz grains may be considered as erratics.

The cement between the tightly fitting quartz grains was originally a very fine parting of limonite, (2→5 μ thick) which because of secondary silicification has retreated behind later diagenetic quartz grain boundaries in many places. Fine subangular quartz grains (approx. 30 μ) as groups between the larger grains (0.1 mm and greater) only locally appear to be absorbed into the larger neighbouring quartz grains by encroaching silicification. No cherts are seen but only occasional quartz rock fragments, some of which contain trains of fluid inclusions.

The rounding of subangular quartz grains by silicification, moderate to poor sorting, with the presence of some erratic grains at 'bottom' of the section, scarcity of both ferruginous cherts and heavy minerals are typical of Pandurra Formation rocks generally.

P 49 : Silcreted gravel or conglomerate

Purple, pink, white and colourless, rounded to subangular, quartzose pebbles and flattened purple shale grains from a fraction of a millimetre to several millimetres in size in a white silica matrix.

Many of the larger grains of quartz mineral, chert grains (clear and ferruginous) and rock fragments (0.7 to 3.0 mm) are quite spherical and similar to those seen in Whyalla Sandstone. Other grains, (about half of larger grains, i.e. + 0.5 mm) are subangular. One large quartz rock fragment consists of cemented grains many of which are near spherical (0.2 to 1.0 mm), the composite grain being subangular and 10 mm maximum diameter. One large subangular 2.5 mm grain of supposed silicified variolitic basalt is seen. About 3 volume percent of the rock consists of ferruginous cherts of very variable size and degree of roundness, as commonly seen in Whyalla Sandstone are present.

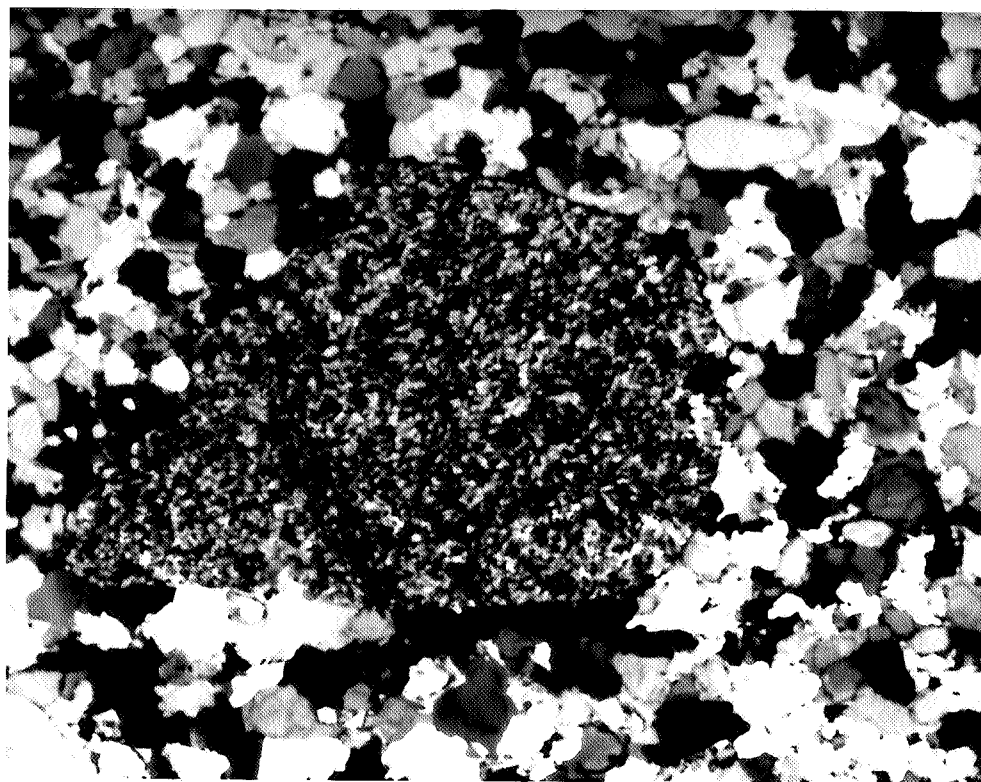


Figure 5

Whyalla Sandstone with Pandurra Quartzite pebble inclusion. P49A.

The 'erratic' (centre of photograph and similar to other erratics in the section) is cherty and cannot positively be identified as a fragment of Pandurra Quartzite.

Crossed nicols

Magnification X 14

The fine matrix of 20 volume percent of rock consists of small angular quartz grains (approx. 0.08 mm) of fairly even grain which are very similar to the silt-size matrix as seen in many Whyalla Sandstone specimens. The rock consists of much derived Whyalla Sandstone material with equally abundant coarse rock material derived from other (Pandurra Formation ?) sources. A secondary silicified gravel derived from several sedimentary sources some of which are noted above.

P 49A : Whyalla Sandstone silicified and containing Pandurra Pebbles (Figure 5).

A fine even-grained pink quartzite with large pebble erratics.

In thin-section the quartz grains are not so even-grained as they appear in hand specimen. However many grains show some internal rounding of a nucleus demarcated by limonite particle inclusions before silicification set-in. Grain size of mainly quartz mineral grains varies between 0.1 and 1.3 mm with median size around 0.4 mm. Overgrowths of quartz have produced an almost entirely quartz mosaic, the originally rounded nature of grains being shown by included fine particles as described above. The typical ferruginous cherts of Whyalla Sandstone and some clear chert grains are rounded to subangular and of median quartz grain size. Some recrystallisation of part of chert is noted in some cases. The cherts occur to about 2 percent of microscope field. Brownish, rounded tourmaline grains (approx. 0.048 mm) occur as inclusions in quartz overgrowths.

Large erratics of 'Pandurra Quartzite' in the rock are rounded, patchily faintly ferruginous recrystallised chert, 20 mm and 3 mm in size; also a more ferruginous fine chert grain is somewhat subangular and 6 mm in size.

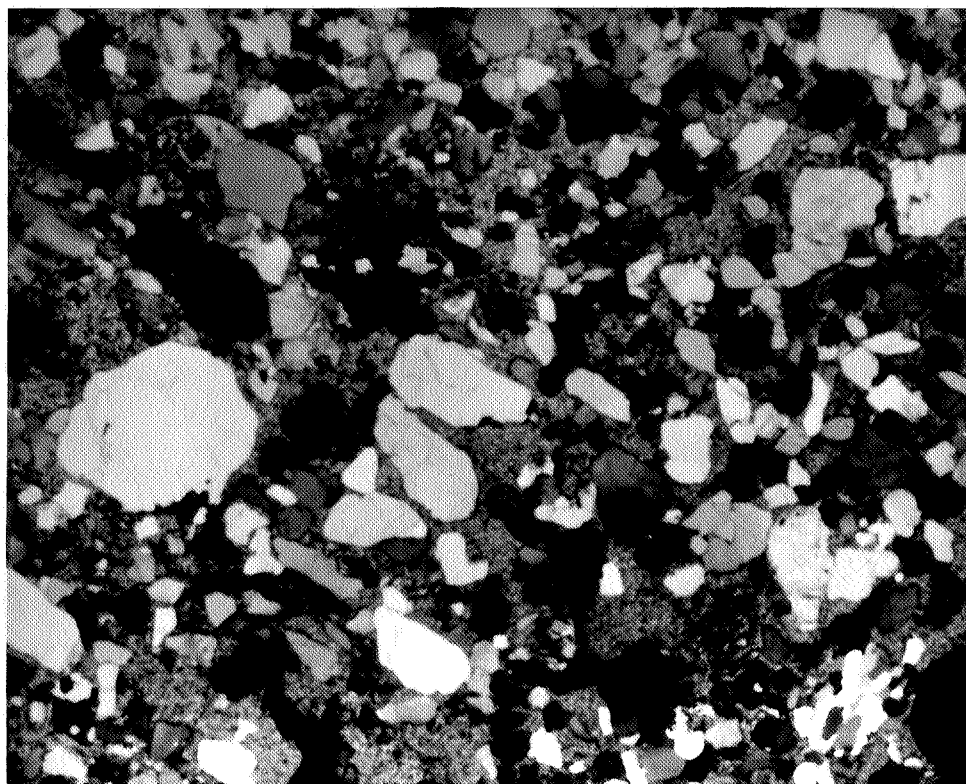


Figure 6

Whyalla Sandstone. P50B...

Quartz (flat white and various shades of grey colour)
occurs in roughly two population distributions
of large and small grains.

Cherts (speckled or mottled black and white).

Clay minerals as matrix (fine stipple).

Crossed nicols

Magnification X 14

A silicified but sharp boundary between the erratics and main quartzite is noted.

Evidence for identification of the rock as Whyalla Sandstone, which has since been silicified, are the original roundness of quartz grains, the relatively even grained size of quartz, and the presence of ferruginous cherts of similar grain size to the quartz grains. The large erratics are not obviously derived from the Pandurra Formation but this is the most likely source.

P 50B : Looks like Pandurra Formation, outcrops like Whyalla Sandstone.

In hand-specimen are seen very irregular quartz grains as subangular to angular grains, abundant clay matrix, local oxide copper (probably traces of malachite).

In thin-section the quartz grains are seen to be not so irregular in grain shapes or sizes. Many grains of quartz rock and quartz mineral are perfectly spherical to near spherical or flat-rounded; this comprises rather more than half the grains present, both small and large. Grain size variation is 0.2 to 3.5 mm. There are, as in normal Whyalla Sandstone, roughly two populations of quartz grains; small, evenly subangular and rounded varying between 0.2 and 0.5 mm, median 0.3 mm; large, 1 to 3.5 mm, some rounded but many rounded-subangular or flattened in shape with a ratio 2:3 rock fragments to mineral grains and comprising about 5 percent of the rock. The larger grains do show a tendency to concentrate in some layers (bedding planes). Ferruginous cherts, and some clear chert grains, subangular to rounded occur to 1 to 2 volume percent of the rock averaging, 0.3 mm and are present interstitial with the finer quartz grains.

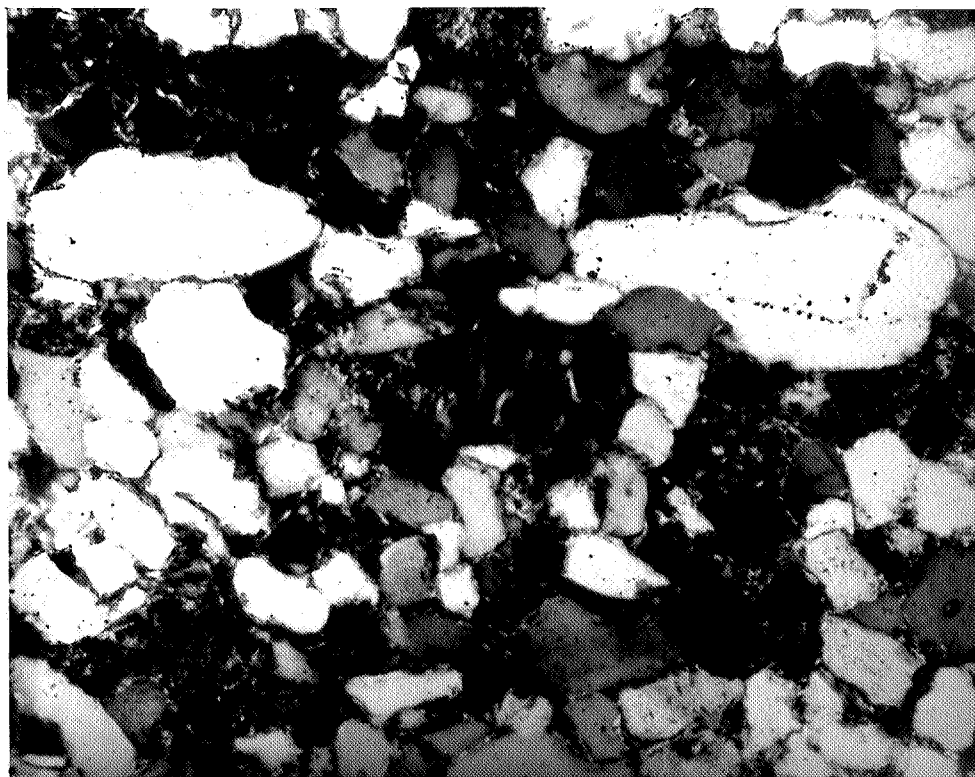


Figure 7

Corruburra Sandstone. P51

Many grains of quartz have a flat shape with rounded corners.

Micaceous cherts, dark or flecked/speckled black and white grains (See centre of photograph).

Illite (white and grey flakes) form fine matrix of narrow channels between quartz and chert grains.

Crossed nicols

Magnification X 105

'Volcanic' (devitrified glass) sericitic cherts are present but are rare. Some replacement of quartz by opaque mineral is noted.

The matrix is of mainly fine illite with some coarse sericitic shreds and kaolinite patches overall to 25 volume percent of the rock. The kaolinite patches may be the result of clay mineral replacement of chert.

The evidence of a degree of roundness of many quartz grains, evidence of two quartz population distributions, presence of appreciable ferruginous cherts and rare volcanic fragments would suggest this to be Whyalla Sandstone.

- P 51 : Sandstone, (Whyalla or Corraberra ?) from Tent Hill crater. Pale sandstone shows a sharp contact between fine and medium grained sandstone (Figure 7).

An even grained sandstone of rounded (not spherical) subangular and flattened quartz grains varying between 0.048 to 0.5 mm, median 0.1 mm. Many grains show strain with Böhm lamellae and most grains are quartz mineral grains. Micaceous cherts are more common than ferruginous cherts, are of similar grain size to median quartz and are rounded to subangular to 5 volume percent of the rock. Clay mineral occurs up to 3 volume percent as interstitial mineral in narrow channels between grains, but the rock cement is to a great extent silicified, though sutured boundaries between quartz grains are rare. Traces of glauconite i.e. clay, as rounded speckled green grains are seen, also occasional large (approx. 0.1 mm) rounded green tourmaline grains with yellow-green overgrowths. Aggregates of leucoxene and some magnetite are seen distributed throughout to a total of 3 volume percent.

Overgrowths on quartz grains appear to be slight, following evenly the original quartz grain margins. These are easily detectable by opaque particle inclusions which indicate original grain shape.

Occasional microcline grains are seen of oblong grain shape and size ranges are similar to those of the quartz grains.

The abundance of strained quartz and of flat-round rather than spherical to near-spherical quartz grains, of micaceous cherts and appreciable leucoxene suggest this rock is a member of the Corraberra Sandstone rock unit.

P.51B : Pandurra Formation. Pink quartzite. Boulder in Whyalla Sandstone.

Pink irregular quartzite with some pale erratic grains. Patchy clay interstices but mainly silicified. Traces of oxide copper mineralisation.

In thin-section the rock is seen to be a poorly sorted quartzite of mainly quartz mineral grains, minor quartz rock fragments (including quartz gneiss fragments) and some clear cherts, the latter grains occurring to around 2 volume percent. Predominantly large subangular and with fewer rounded quartz mineral grains and rock fragments (approx. 2 mm in size) occur to about 3 volume percent of the rock. The remaining grains are mainly quartz mineral grains from 0.1 to 1.2 mm. median 0.3 mm and a poorly-sorted banding of mainly fine grains with large erratics and alternatively mainly coarse with much interstitial fine quartz are seen. The rock consists largely of fused sutured quartz mosaics although cherty material in the interstices and in vugs and a little clay mineral in blind channels do occur.

Heavy minerals and feldspar are lacking.

The poor sorting, lack of ferruginous cherts, subangularity of most grains and lack of heavy minerals favour classifying the rock as Pandurra Quartzite.

P 52 : ? Woocalla Dolomite. Manganiferous.

Fine cream-coloured rock with numerous small vugs. Manganese dioxide has apparently deposited (?) along sub-parallel fractures or joints in the rocks and has diffused along microfractures flanking the major joints.

Thin-section shows fine-grain dolomite with moderate (3 volume percent) goethite depositing largely as spheres (approx. 5μ) within fine dolomite grains. Possibly some of the goethite is pseudomorphing original syngenetic pyrite (?). The dolomite which is seen as irregular grains of very variable size (to 0.026 mm) is fairly clear except where deposited manganese dioxide occurs in the dolomite as irregular lenticles and masses commonly intergrown with goethite although as mentioned above, most goethite occurs separately scattered through the dolomite.

It would appear that both MnO_2 and goethite were the product of weathering of dolomite containing appreciable iron and manganese in solid solution. Upon weathering both elements were leached and became oxidised, the iron precipitating first in the dolomite followed by manganese which precipitated in fractures. Thus the diffuse goethite seen beyond and around manganese in fractures is precipitated iron extracted from dolomite; but the goethite spheres which occur scattered throughout the rock may possibly be pyrite pseudomorphs.

No organic algal structures are seen.

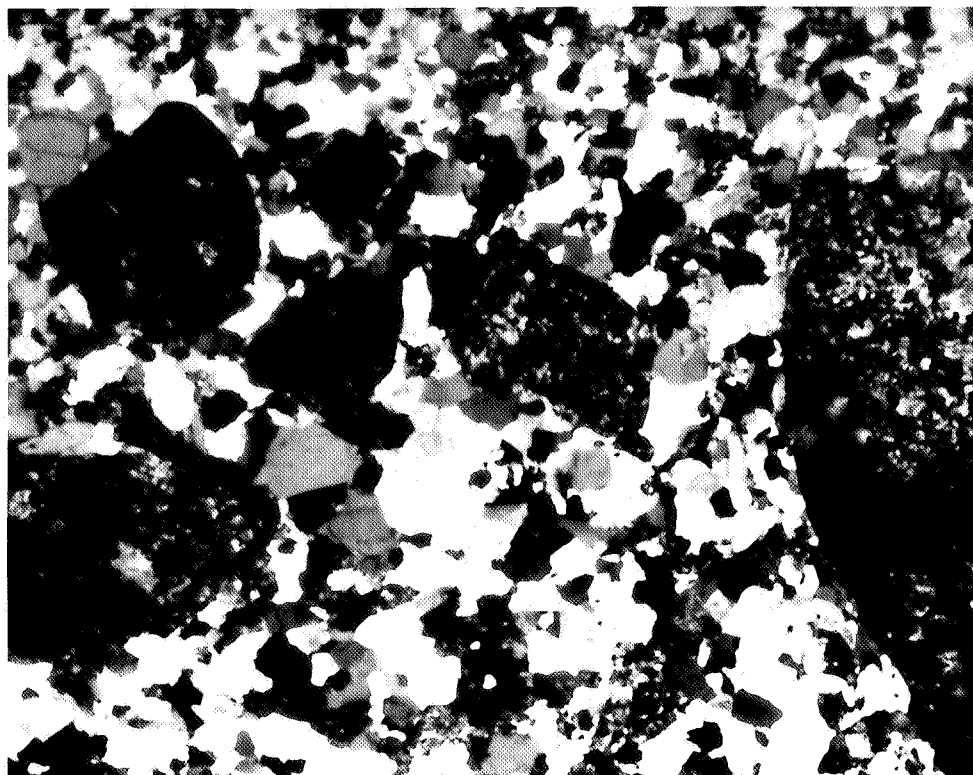


Figure 8

Whyalla Sandstone with abundant volcanic clasts. P53

Quartz of Whyalla Sandstone (flat grey and white colours) often shows primary rounded centres with subangular quartz overgrowths which together form mosaics.

Volcanic clasts are mainly rounded devitrified glass as cherty masses (speckled black and white or mainly dark masses in photograph).

Crossed nicols

Magnification X 16

Woocalla Dolomite where weathered, quite often shows oxidation of 'dolomite' manganese, although as previously seen by the writer it occurs more diffuse associated with goethite and seems to occur towards the bottom of the stratigraphic unit.

P 53 : Whyalla Sandstone containing chips of Gawler Volcanics (Figure 8)

A spongy looking pinkish quartzite with irregular inclusions of limonitised fine-grained rock.

In thin-section the quartz grains are presently irregular in size and subangular due in some cases to quartz overgrowths on more rounded grains. The size variation is 0.1 to 1.3 mm, median 0.2 mm. Some large grains of quartz are quite subangular and this is fairly unusual for Whyalla Sandstone as seen by the writer. However most grains (>80 volume percent of quartz grains present) which now have quartz overgrowths, show by internal narrow elliptical trains of hematite particles that they were to some extent rounded before silicification. The quartz grains which were originally mainly mineral grains constitute 85 volume percent of the rock and with associated volcanic clasts, form a tight silicified mosaic. The remaining 15 percent of the rock consists of variable size (to several mms) and variable shape ferruginous cherts, clear cherts, micaceous cherts and cherty grains with micro-lites (variolitic) texture and devitrified palagonite always accompanied by abundant limonite. Other grains contain chertified feldspar phenocrysts in a devitrified glassy mass of porphyrite or andesite? The latter are derived fragments from the Gawler Volcanics. Less than 1 percent of rounded zircons (approx. 0.048 mm in size) are present.

A poorly sorted quartzite with some rounding shown by primary nuclei of quartz grains. However the quartzite fabric is somewhat similar to that seen in A4255 (DH 3-4-3D, 6.1 m), Whyalla Sandstone from the Main Open Cut area (Ref. 5). This rock has been highly silicified but does not contain as much cherty material.

A Whyalla Sandstone with abundant volcanic clasts.

P 56 : Whyalla Sandstone, silicified.

Mainly rounded to near spherical quartz grains showing frosted and some grooved surfaces with a few large angular grains and many ferruginous cherty grains, all in a white fine-grained interstitial matrix.

The Whyalla Sandstone quartz grains in thin-section are mainly mineral grains (approx. only 1 percent quartz rock fragments are present) and are quite spherical, varying between 0.2 and 1.2 mm, median 0.4 mm, associated with subangular to rounded ferruginous volcanic chert grains of similar size range occurring to 10 volume percent of the rock. The interstices (5 volume percent of rock) consists of fractured chert fragments cemented by secondary quartz (also chert) associated with some clay mineral to neighbouring quartz grains. Many of the ferruginous chert grains contain devitrified glassy matter, some variolitic structures and are possibly basaltic. One large round greenish tourmaline (0.4 mm) shows fracture and fill with chert. Opaque oxides occur as magnetite grains in groups inside the volcanic cherts and occasionally within quartz overgrowths on primary quartz grains.

Whyalla Sandstone with abundant volcanic cherty clasts and cherty/clay interstitial material which is mainly in vugs.

- P 57A : Silicified conglomerate; overlies Pandurra Formation. Believed to have undergone pre-Marinoan silicification along with upper section of the Pandurra Formation.

A silicified 'conglomerate' of large and small grains in two distinct population distributions.

Thin-section shows large round quartz rock fragments and chert from 1 to 7 mms (about 20 volume percent of the rock) in a more subangular matrix of smaller quartz grains with fewer rock fragments and clear chert grains (overall 0.05 to 0.6 mm, median 0.3 mm, many grains are of approximately median size). The small quartz grains show the usual elliptical trains of hematite particles just within the grain boundaries to indicate more rounded primary quartz. Secondary silicification has welded the whole mass of quartz grains into sutured quartz mosaics. Occasional (< 1 volume percent of rock) volcanic rock grains of devitrified glass containing decussate patterns of micro-lites are observed. The large round conglomerate fragments which are quartz rock fragments show a similar complete silicification of constituent grains into mosaics.

The rudaceous element is largely of water-worn quartz and some chert grains derived from the Pandurra Quartzite. Absence of heavy minerals, ferruginous chert and aeolian quartz would suggest a Pandurra Formation origin for much of the rock material. A pebbly sand (Ref. 3) type of conglomerate which was subsequently silicified.

- P 58 : Pandurra Formation grit. Unsilicified.

A poorly sorted sandstone of mainly subangular quartz grains and rock fragments. Clay matrix.

Thin-section shows many of the larger quartz grains to be rounded and are commonly perfectly oval in section. These grains are mainly quartz mineral. The quartz mineral grains, quartz rock fragments and clear chert are of similar size (1.5 to 2 mm). Many of the larger quartz mineral grains have rounded primary nuclei and silicification has made the grains more angular.

The smaller grains in the rock vary mostly from 0.1 to 1.0 mm, are too variable to obtain median value; they are more rounded (some near spherical) than subangular.

Overall, the modal volume percent of mineral constituents are :-

Quartz mineral grains	90
Quartz rock fragments	5
Clear chert clasts (large grains)	2
Ferruginous rounded volcanic chert grains (small grains)	1
Kaolinite matrix	2

The rock is mainly silicified into mosaics and clay mineral occurs only in vugs.

The rock bears a passable resemblance to Whyalla Sandstone A4255 (Open Cut) in original shapes of quartz grains before and after silicification to produce silicified mosaics. However compared with other thin-sections of Whyalla Sandstone from the Gunson area but principally from the Pandurra area the grains are still overall mainly subangular or (~ 80 volume percent of grains present) taking into consideration quartz overgrowths, and the ferruginous cherts are present in very minor quantities.

The rock is silicified Pandurra Formation but is a borderline transition type to Whyalla Sandstone.

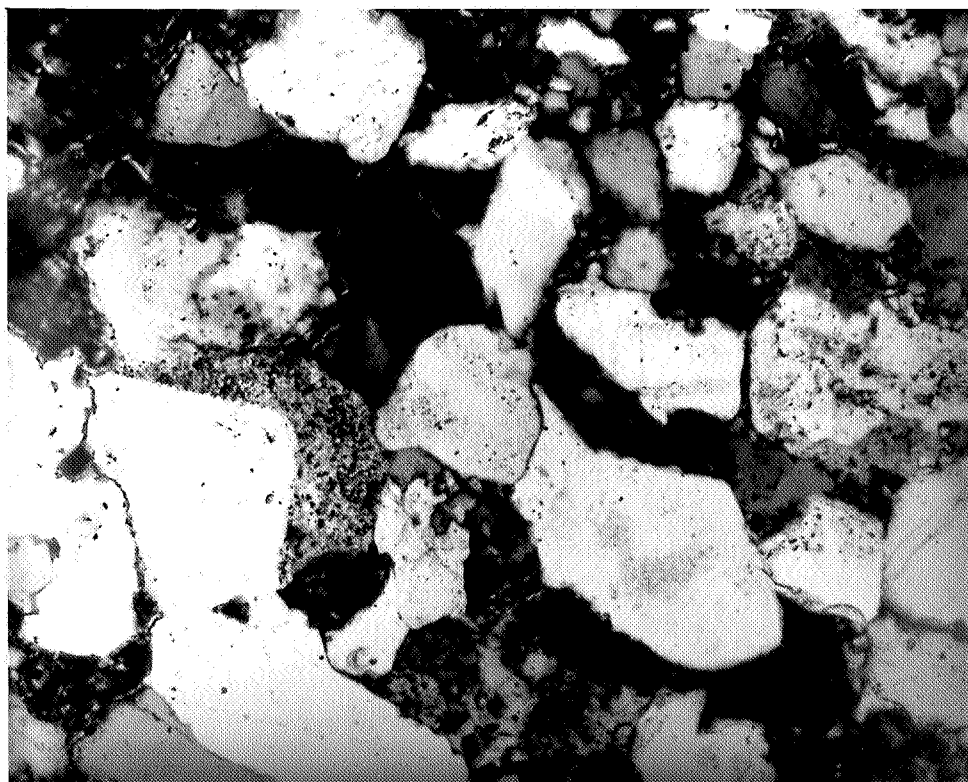


Figure 9

Whyalla Sandstone - silicified P60

Many subangular quartz grains are present but
also appreciable rounded grains remain.
Few silicified grains show opaque or fluid
inclusion trains.

Cherts are grey and white and stippled black
and white (See centre near bottom margin
and near top centre of photograph).

Interstitial grey and white flakes are kaolinite,
speckled grey and white are chert.

Crossed nicols

Magnification X 101

P 60 : Whyalla Sandstone. Silicified lithic sandstone
(Figure 9)

Many rounded quartz grains with frosted surfaces in a matrix of finer more angular grains and clay.

Thin-section shows banding by the larger quartz and chert (ferruginous and clear) grains which are rounded (near spherical in some grains) or are flattened grains with rounded corners (0.2 to 1.5 mm) with interstitial finer quartz grains (0.02 to 0.19 mm). These bands alternate with those of predominantly small subangular quartz grains (also 0.02 to 0.19 mm) with scattered large erratics of round larger quartz grains. Other bands of equigranular quartz (0.2 to 0.4 mm) many rounded, few subangular grains occur between the coarse and fine bands described. Some rough sorting of grains has therefore taken place. The overall percentage of oval to near spherical grains in the section is probably around 20 volume percent but is much higher among the bands of largest quartz grains. The overall percentage of ferruginous chert grains is around 2 volume percent with some grains consisting of devitrified palagonite glass. Some rounded topaz and less brownish and green tourmaline occur, both minerals are around 0.08 mm in size and with the occurrence of sporadic opaque minerals of similar size make up about 1 volume percent of the rock. The tourmaline is quite angular and appears to have suffered very little attrition. The rock has been largely silicified into mosaics but there are blind interstitial channels containing chert and kaolinite.

A silicified non-standard Whyalla Sandstone. The abundance of volcanic cherts and spherical large quartz grains puts the identity beyond doubt.

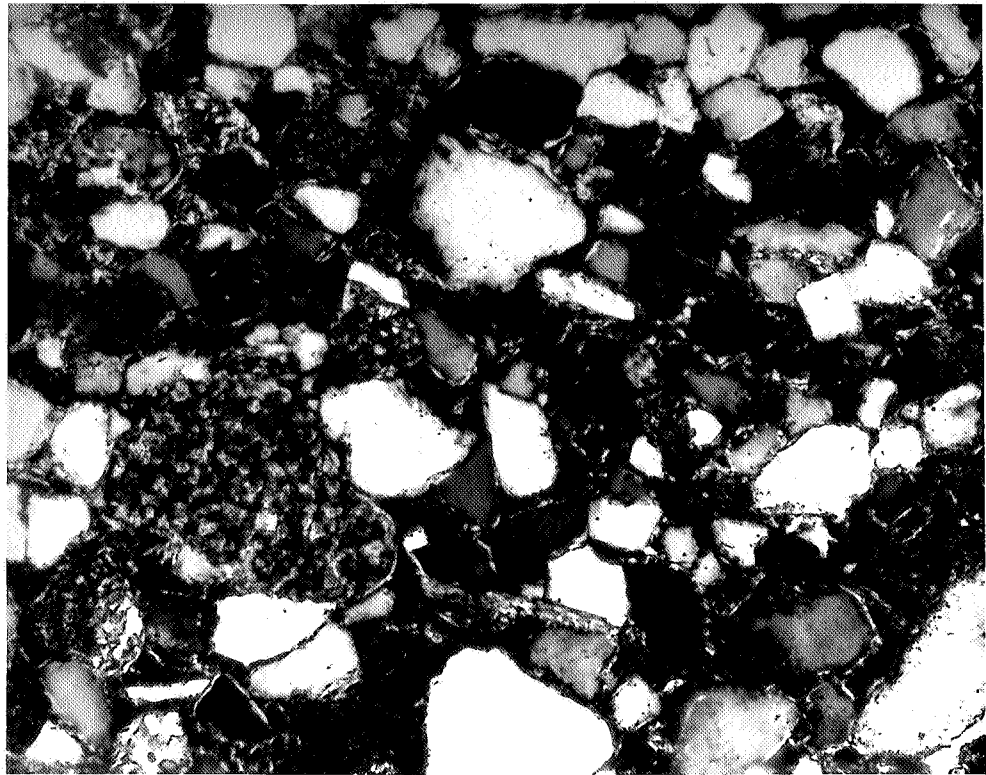


Figure 10

Corruburra Sandstone with much derived Whyalla Sandstone material. P65

Many quartz grains (flat white and grey colours) show a fair degree of rounding and quite a number are of somewhat flattened appearance.

Volcanic cherty fragments (left of centre); clear chert (near left bottom corner).

Otherwise opaque leucoxene and flakey mica in interstices. Heavy minerals other than leucoxene are present but do not show up on photograph.

Crossed nicols

Magnification X 103

P 61 : Silcrete contains fibrous laths.

Rounded quartz grains in a fine white matrix but large clasts of glassy volcanic material also occurs.

The quartzite consists largely of rounded quartz grains, even polygonal grains (stress relief during silicification ?), quartz replaced cherts and some ferruginous chert grains, all grains being between 0.16 and 0.80 mm, median 0.3 mm. Internal opaque mineral trains show that original quartz grains before silicification were very much more angular than they are now. Where not welded together at mutual boundaries, fine slivers of chert are seen to be part of the rock cement.

Ferruginous banded chert (several millimetres in length and width) form most of the large clasts. However, one clast is of quartzite with similar constituent quartz grain sizes to those in the matrix quartzite but having a ferruginous cherty cement between the quartz grains and also containing a few ferruginous and clear chert grains of similar grain size to the constituent quartz grains. One volcanic chert several millimetres in size contains chert and ferruginous matter which has filled amygdules (?) and replaced feldspar (?) laths; all in a ferruginous cherty matrix (Gawler Volcanics ?).

A recrystallised quartzite containing cherty sediments and volcanic fragments.

Fibrous laths are of fine ferruginous cherty material.

P 65 : ? Whyalla Sandstone. Clay-rich layers similar to P40, P 41A. Fine white quartzite with round lithic fragments as inclusions. (Figure 10).

Rounded, flat with rounded corners and subangular grains 0.03 to 1 mm, median 0.08 mm are poorly sorted and

are mainly quartz mineral grains with appreciable ferruginous cherts. The flattened grains among both the smaller and larger grains are present to nearly half the total quartz grains in the section. Ferruginous cherts and minor granophyric grains which are similar in shape are together present to less than 1 volume percent. Clear cherts of all sizes are rounded and occur to less than 1 percent. Heavy minerals including leucoxene> magnetite> zircon> tourmaline> rutile are present, total between 1 and 2 percent. Also scattered large clasts (0.3 to 1 mm) of spherical quartz rock fragments occur to 3 percent of grains present in most of section but grades up to 10 percent where present in a coarser band of quartz grains.

Clay mineral (kaolinite> illite) as fine flakes to 10 volume percent of the rock, some coarser muscovite also occurs to less than 1 percent of rock, also a little interstitial carbonate.

The rock is similar to P 40 in grain size and texture but contains no feldspar, less ferruginous chert, more clay and more heavy minerals.

A Corraberra Sandstone with some derived Whyalla Sandstone material (near spherical larger quartz grains and ferruginous lithic chert fragments).

P 65A : Whyalla Sandstone ? Similar to P 65.

A fine-grain greenish equigranular sandstone specked with red lithic particles and white clay and black magnetite (?).

Overall, coarser grained than P 40, 41A and 65, (has many fewer fine grains and is more even grained than the latter specimens).

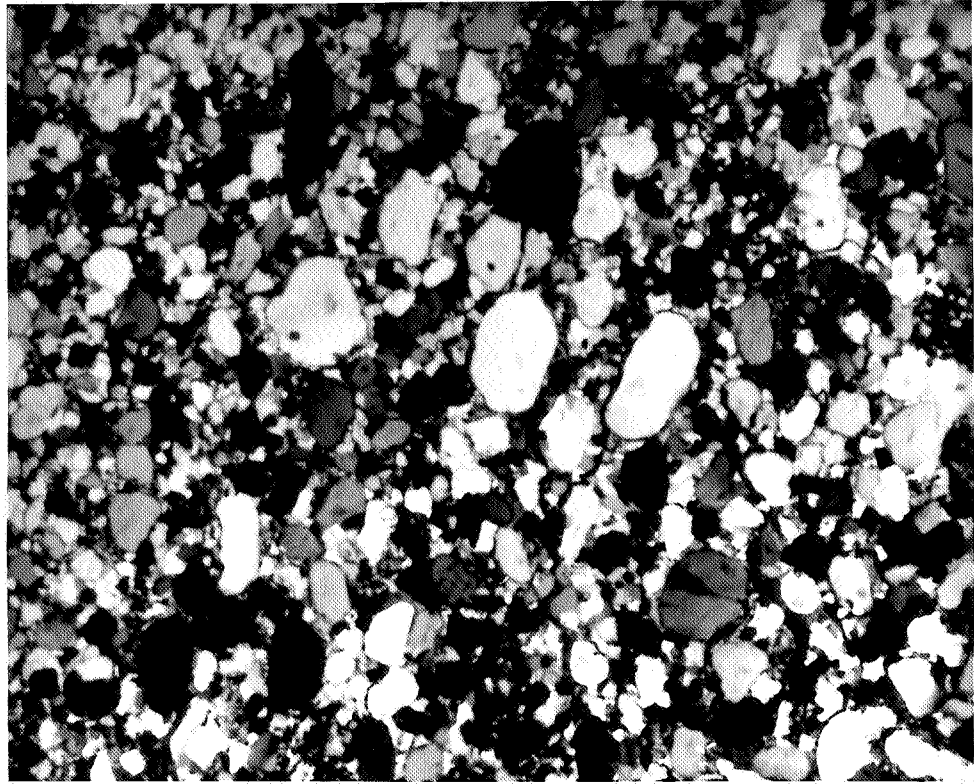


Figure 11

Corraberra Sandstone with derived Whyalla Sandstone material - P66

Larger quartz grains tend to be somewhat rounded or flat/round with smaller interstitial quartz particles being subangular.

Fine clay mineral (black and white flecked) is a film between quartz grains or is in vugs. Magnetite (black). Other heavy minerals though present are not distinguishable at this magnification.

Crossed nicols

Magnification X 14

Many quartz grains are round, near spherical, some are subangular and fewer are flattened with round corners. The grains vary between 0.16 and 0.80 mm, but most grains are around the median of 0.3 mm. Similarly quartz rock fragments, some being gneissose occur to 3 volume percent of the rock. Rounded ferruginous cherts, some containing silicified volcanic microlites and occasionally included mica flakes occur to 2 to 3 volume percent and also clear chert grains of similar size to 1 percent of rock. Clay mineral as mainly illite with less kaolinite occupies vugs and narrow channelways between quartz grains. Total clay mineral is about 3 volume percent of rock.

Occasional rounded greenish tourmaline grains to much less than 1 volume percent are the only heavy minerals seen. Some silicification of quartz grains into mosaics has taken place.

There is some resemblance in quartz grain shapes to Corraberra Sandstone of Mt. Gunson, (A4223), but a greater abundance of cherty lithic grains and fewer heavy minerals are observed. In shapes of grains with a high degree of rounding, lack of feldspar, the presence of abundant volcanic chert (though finer Corraberra Sandstone often contains an abundance of this lithological type) and the presence of only minor heavy minerals as tourmaline, the resemblance is therefore greater to Whyalla Sandstone. Possibly a transition rock type?

P66 : Sandstone (Whyalla or Corraberra). (Figure 11).

Inequigranular, rounded or rounded-flat grains and abundant lithic particles. A white sandstone as per P 65.

A quartzite with two population distributions of larger near spherical grains and smaller round to subangular grains. Some of the larger grains do show a tendency to flatness but these are concave-convex and concave shapes as

are commonly seen in some large Whyalla Sandstone quartz grains. Grain size of larger quartz grains is 0.3 to 1.0 mm, smaller grains vary between 0.03 to 0.15 mm, median 0.1 mm. Ferruginous chert varies from subangular to rounded-flat grains, vary between 0.05 and 0.5 mm and occur to 1 to 2 volume percent. Clear cherts are not seen. A brownish clay mineral occurs between 7 and 10 volume percent of the section (derived from volcanic material?) and is seen occasionally in small vugs but mainly as a film between the quartz and chert grains. Magnetite occurs to 1 volume percent; rounded zircon > rutile > tourmaline also makes up to 1 percent.

Not so differentiated into two quartz size populations as A4247 (Whyalla Sandstone, Mystery area) but similar degree of rounding of large quartz grains and similar ferruginous chert content. Somewhat similar to P 41A, Corraberra Sandstone in quartz grain sizes and abundance and variety of heavy minerals. Probably Corraberra Sandstone with derived Whyalla Sandstone material.

P 67 : Sandstone - please identify.

An irregular sandstone of large subangular quartz grains in a fine subangular quartz grain matrix.

At one end of the section are the familiar (of Pandurra Quartzite) large angular to subangular erratics of quartz rock fragments which are more abundant than large mineral grains, (varying 0.5 to 4 mm) in fine quartz (approx. 0.1 mm). Further down the section are subangular to round quartz grains (0.3 to 0.8 mm, mostly around median 0.7 mm) which show peripheral trains of opaque inclusions indicating secondary silicification. Silicification has not altered the primary shapes of quartz grains very much. Where not silicified, and this is the case in areas all over the section, limonite fills in spaces between the quartz grains to about 2 volume percent of the rock. Clay mineral as

000228

kaolinite occupies sporadic vugs in the rock and is seen adjoining the limonite in the interstices; also elsewhere brownish cherty material is seen associated with kaolinite in the interstices. No chert grains, ferruginous or clear are seen and no heavy minerals apart from occasional almost fresh unworn zircon grains. Absence of feldspar, ferruginous cherts and near absence of heavy minerals and the angularity of large quartz grains suggest this rock to be a member of the Pandurra Formation. The quartz grain fabric is somewhat similar to specimens A4252, 4254 and 4256 from the Main Open Cut at Mount Gunson.

4. REFERENCES

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